

7.8 Action Tables

The following tables provide a quick reference on user actions provided by Ozone (see *User Actions* on page 42).

7.8.1 Breakpoint Actions

Actions that modify the debugger's breakpoint state.

Action	Description
Break.Clear	Clears an instruction breakpoint.
Break.ClearOnSrc	Clears a source breakpoint.
Break.ClearOnData	Clears a data breakpoint.
Break.ClearOnSymbol	Clears a data breakpoint on a symbol.
Break.ClearAllOnData	Clears all data breakpoints.
Break.ClearAll	Clears all code breakpoints.
Break.Disable	Disables an instruction breakpoint.
Break.DisableOnSrc	Disables a source breakpoint.
Break.DisableOnData	Disables a data breakpoint.
Break.DisableOnSymbol	Disables a data breakpoint on a symbol.
Break.Enable	Enables an instruction breakpoint.
Break.EnableOnSrc	Enables a source breakpoint.
Break.EnableOnData	Enables a data breakpoint.
Break.EnableOnSymbol	Enables a data breakpoint on a symbol.
Break.Edit	Edits a breakpoints advanced properties.
Break.EditOnData	Edits a data breakpoint.
Break.EditOnSymbol	Edits a data breakpoint on a symbol.
Break.OnChange	Sets a data breakpoint on a symbol.
Break.Set	Sets an instruction breakpoint.
Break.SetEx	Sets an instruction breakpoint.
Break.SetOnSrc	Sets a source breakpoint.
Break.SetOnSrcEx	Sets a source breakpoint.
Break.SetType	Sets a breakpoint's implementation type.
Break.SetCommand	Assigns a script callback function to a breakpoint.
Break.SetCmdOnAddr	Assigns a script callback function to a breakpoint.
Break.SetOnData	Sets a data breakpoint.
Break.SetOnSymbol	Sets a data breakpoint on a symbol.
Break.SetVectorCatch	Edits the vector catch state.

7.8.2 Code Profile Actions

Code profile related actions.

Action	Description
Coverage.Exclude	Filters program entities from the code coverage statistic.
Coverage.Include	Re-adds program entities to the code coverage statistic.
Coverage.ExcludeNOPs	Filters trailing NOPs from the code coverage statistic.

Action	Description
Profile.Exclude	Filters program entities from the code profile statistic.
Profile.Include	Re-adds program entities to the code profile statistic.
Profile.Reset	Clears code profile data and resets all execution counters.

7.8.3 Debug Actions

Actions that modify the program execution point and that configure the debugger's connection, reset and stepping behavior.

Action	Description
Debug.Connect	Connects the debugger to the target.
Debug.Continue	Resumes program execution.
Debug.Disconnect	Disconnects the debugger from the target.
Debug.Download	Downloads the program file to the target.
Debug.Halt	Halts program execution.
Debug.IsHalted	Queries the program state.
Debug.Reset	Reset the program.
Debug.ReadIntoInstCache	Reads a code block into the instruction cache.
Debug.RunTo	Advances program execution to a particular location.
Debug.SetConnectMode	Sets the connection mode.
Debug.Start	Starts the debug session.
Debug.Stop	Stops the debug session.
Debug.StepInto	Steps into the current function.
Debug.StepOver	Steps over the current function.
Debug.StepOut	Steps out of the current function.
Debug.SetNextPC	Sets the next machine instruction to be executed.
Debug.SetNextStatement	Sets the next source statement to be executed.
Debug.SetResetMode	Sets the reset mode.
Debug.SaveSnapshot	Saves a debug snapshot.
Debug.LoadSnapshot	Loads a debug snapshot.

7.8.4 Edit Actions

Actions that edit behavioral and appearance settings of the debugger.

Action	Description
Edit.Color	Edits an application color.
Edit.DisplayFormat	Edits an item's integer value display format.
Edit.Font	Edits an application font.
Edit.MemZone	Edits the memory zone of a watched expression.
Edit.Preference	Edits a user preference.
Edit.RefreshRate	Edits the refresh rate of a window or watched expression.
Edit.SysVar	Edits a system variable.

7.8.5 ELF Actions

Actions for retrieving ELF program file information.

Action	Description
Elf.GetBaseAddr	Returns the program file's download address.
Elf.GetFileClass	Returns the ELF file class of the program file.
Elf.GetEntryPointPC	Returns the initial value of the program counter.
Elf.GetEntryFuncPC	Returns the first PC of the program's entry function.
Elf.GetExprValue	Evaluates a symbol expression.
Elf.GetEndianness	Returns the program file's byte order.
Elf.SetConfig	Configures the ELF parser.
Elf.PrintSectionInfo	Prints ELF file section information.

7.8.6 Export Actions

Actions that export debug session data to CSV or text files.

Action	Description
Export.CodeProfile	Exports code profile data.
Export.DataGraphs	Exports all data graphs to a CSV file.
Export.Disassembly	Exports program disassembly.
Export.PowerGraphs	Exports all power graphs to a CSV file.
Export.Trace	Exports instruction trace data to a CSV file.

7.8.7 File Actions

Actions that perform file system and related operations.

Action	Description
File.Close	Closes a source code document.
File.CloseAll	Closes all open source code documents.
File.CloseAllButThis	Closes all but the active source code document.
File.CloseAllUnedited	Closes all unedited documents.
File.Exit	Closes the application.
File.Find	Searches for a text pattern.
File.Load	Loads a file.
File.NewProject	Creates a new project.
File.NewProjectWizard	Opens the Project Wizard.
File.Open	Opens a file.
File.OpenRecent	Reopens a recently opened program file.
File.OpenProjectInEditor	Opens the project file within the source viewer.
File.Reload	Reloads a file from disk.
File.SaveProjectAs	Saves the project file under a new file path to disk.
File.Save	Saves an open document to disk.
File.SaveAs	Saves an open document under a new file path to disk.
File.SaveCopyAs	Saves a copy of an open document to disk.

Action	Description
File.SaveAll	Saves all modified files.
File.SelectInExplorer	Selects a source file within the file explorer.

7.8.8 Find Actions

Actions that locate program entities.

Action	Description
Find.Function	Locates a program function.
Find.GlobalData	Locates a global symbol.
Find.SourceFile	Locates a source file.
Find.Text	Opens the Quick Find Widget.
Find.TextInFiles	Opens the Find In Files Dialog.
Find.TextInTrace	Opens the Find In Trace Dialog.

7.8.9 Help Actions

Actions that display help related information.

Action	Description
Help.About	Shows the About Dialog.
Help.Commands	Prints the command help to the Console Window.
Help.UserGuide	Displays the user guide and reference manual.
Help.ReleaseNotes	Displays the release notes.
Help.LicenseManager	Opens the license manager.

7.8.10 J-Link Actions

Actions that perform J-Link operations.

Action	Description
Exec.AddCommandOnOpen	Schedules a J-Link command to be executed immediately before or after opening J-Link connection (i.e. <code>JLink_Open()</code> is called).
Exec.Connect	Connects the debugger to the target.
Exec.Command	Executes a J-Link/J-Trace command.
Exec.Download	Downloads a program or a data file to target memory.
Exec.Reset	Performs a hardware reset of the target.

7.8.11 OS Actions

Actions that perform RTOS related operations.

Action	Description
OS.AddContextSwitchSymbol	Identifies a code symbol that executes a task switch.

7.8.12 Process Actions

Actions that control interaction with external processes.

Action	Description
Process.Exec	Spawns a process and executes an external application.

7.8.13 Project Actions

Actions that configure the debugger for operation in a particular software and hardware environment.

Action	Description
Project.AddSvdFile	Adds a register set description file.
Project.AddRTTSearchRange	RTT configuration command.
Project.AddFileAlias	Sets a file path alias.
Project.AddPathSubstitute	Replaces substrings within source file paths.
Project.AddRootPath	Specifies the program's root path.
Project.AddSearchPath	Adds a path to the program's list of search paths.
Project.ConfigSWO	Configures the Serial Wire Output (SWO) interface.
Project.ConfigSemihosting	Configures the Semihosting interface.
Project.ConfigDisassembly	Edits disassembler options.
Project.DisableSessionSave	Disables saving of individual session information.
Project.RelocateSymbols	Relocates one or multiple symbols.
Project.SetDevice	Specifies the target device.
Project.SetFlashLoader	Specifies the flash loader(s) to be used for one or more flash banks.
Project.SetHostIF	Specifies the host interface.
Project.SetTargetIF	Specifies the target interface.
Project.SetTIFSpeed	Specifies the target interface speed.
Project.SetJTAGConfig	Configures the JTAG target interface.
Project.SetTraceSource	Selects the trace source to use.
Project.SetTracePortWidth	Specifies the number of trace pins comprising the TP.
Project.SetTraceTiming	Configures the trace pin sampling delays.
Project.SetRTT	Enables or disables Real Time Transfer (RTT).
Project.SetCorePlugin	Specifies the file path of the target support plugin.
Project.SetDisassemblyPlugin	Specifies the disassembly support plugin to be used.
Project.SetSmartViewPlugin	Specifies the SmartView plugin to be used.
Project.SetOSPlugin	Specifies the RTOS awareness plugin to be used.
Project.SetBPTType	Sets the allowed breakpoint implementation type.
Project.SetMemZoneRunning	Sets the default zone accessed when the CPU is running.
Project.SetJLinkScript	Sets the J-Link-Script to be executed on debug start.
Project.SetJLinkLogFile	Sets the text file that receives J-Link/J-Trace logging output. In case of debugging via a GDB server the communication between GDB client and GDB server is written into that file.
Project.SetConsoleLogFile	Sets the text file that receives console window output.
Project.SetTerminalLogFile	Sets the text file that receives terminal window output.

7.8.14 Register Actions

Actions that inform about target register properties.

Action	Description
Register.Addr	Returns the memory location of a target register.

7.8.15 Script Actions

Actions that perform script operations.

Action	Description
Script.DefineConst	Defines an integer constant to be used within the project script.
Script.Exec	Executes a project file script function.

7.8.16 Show Actions

Actions that navigate to particular objects displayed on the graphical user interface.

Action	Description
Show.CallGraph	Displays the call graph of a function.
Show.Data	Displays the data location of a program variable.
Show.Definition	Displays the source code definition location of a symbol.
Show.Declaration	Displays the source code declaration location of a symbol.
Show.Disassembly	Displays the assembly code of an object.
Show.InstTrace	Displays a position in the instruction execution history.
Show.Line	Displays a text line in the active document.
Show.Memory	Displays a memory location.
Show.MemoryMap	Displays a symbol within the memory map of the target.
Show.NextResult	Displays the next search result item.
Show.PC	Displays the PC instruction in the Disassembly Window.
Show.PCLine	Displays the PC line in the Source Viewer.
Show.PrevResult	Displays the previous search result item.
Show.Source	Displays the source code location of an object.
Show.ValueData	Displays the symbol pointed to within the memory window.
Show.ValueDisassembly	Displays the symbol pointed to within the disassembly window.
Show.ValueSource	Displays the symbol pointed to within the source viewer.

7.8.17 Snapshot Actions

Actions to program snapshot operations.

Action	Description
Snapshot.LoadReg	Reads a register from a snapshot and writes it to target.
Snapshot.LoadU32	Reads a memory value from a snapshot and writes it to target.
Snapshot.ReadReg	Reads a register from a snapshot.

Action	Description
Snapshot.ReadU32	Reads a memory value from a snapshot.
Snapshot.SaveReg	Saves a register to a snapshot.
Snapshot.SaveU32	Saves a memory value to a snapshot.

7.8.18 Target Actions

Actions that perform target memory and register IO.

Action	Description
Target.AddMemorySegment	Adds a memory segment to the memory map.
Target.EraseChip	Erases the target's FLASH memory (to 0xFF).
Target.FillMemory	Fills a block of target memory with a particular value.
Target.FillMemoryEx	Fills a block of target memory with a particular value and a particular word width.
Target.GetReg	Reads a target register.
Target.LoadMemory	Downloads the contents of a data file to target memory.
Target.LoadMemoryMap	Loads a memory map from a memory map file.
Target.PowerOn	Toggles target power supply by J-Link/J-Trace.
Target.ReadU32	Reads a word from target memory.
Target.ReadU16	Reads a half word from target memory.
Target.ReadU8	Reads a byte from target memory.
Target.SaveMemory	Saves a block of target memory to a binary data file.
Target.SetAccessWidth	Specifies the memory access width.
Target.SetEndianness	Configures the debugger for a particular data endianness.
Target.SetReg	Writes a target register.
Target.WriteU32	Writes a word to target memory.
Target.WriteU16	Writes a half word to target memory.
Target.WriteU8	Writes a byte to target memory.

7.8.19 Timeline Actions

Actions related to the Timeline Window.

Action	Description
Timeline.Reset	Resets trace and sampling data.

7.8.20 Tools Actions

Actions that open tool dialogs.

Action	Description
Tools.JLinkSettings	Opens the J-Link Settings Dialog.
Tools.Preferences	Opens the User Preference Dialog.
Tools.SemihostingSettings	Opens the Semihosting Settings Dialog.
Tools.SysVars	Opens the System Variable Editor.
Tools.TraceSettings	Opens the Trace Settings Dialog.

7.8.21 Toolbar Actions

Actions that modify the state of toolbars.

Action	Description
Toolbar.Show	Displays a toolbar.
Toolbar.Close	Hides a toolbar.
Toolbar.AddCustomButton	Adds a button to the Custom Toolbar.
Toolbar.RemoveCustomButton	Removes a button from the Custom Toolbar.
Toolbar.EnableCustomButton	Enables a button in the Custom Toolbar.
Toolbar.DisableCustomButton	Disables a button in the Custom Toolbar.
Toolbar.PressButton	Performs the same action as when clicking on a button in a toolbar.

7.8.22 Trace Actions

Trace-related actions.

Action	Description
Trace.SetPoint	Sets a tracepoint.
Trace.ClearPoint	Clears a tracepoint.
Trace.EnablePoint	Enables a tracepoint.
Trace.DisablePoint	Disables a tracepoint.
Trace.ClearAllPoints	Clears all tracepoints.
Trace.Reset	Resets instruction trace data.

7.8.23 Utility Actions

Script function utility actions.

Action	Description
Util.Error	Shows an error message box and stops the debug session.
Util.Log	Prints a message to the console window.
Util.LogHex	Prints a formatted message to the console window.
Util.Sleep	Pauses the current operation for a given amount of time.

7.8.24 Window Actions

Actions that edit the state of debug information windows.

Action	Description
Window.Add	Adds a symbol to a window.
Window.Close	Closes a window.
Window.CloseAll	Closes all windows.
Window.Clear	Clears a window.
Window.CollapseAll	Collapses all items of a window.

Action	Description
Window.ExpandAll	Expands all items of a window.
Window.Export	Exports the window content to file.
Window.Insert	Inserts a symbol into a window.
Window.Remove	Removes a symbol from a window.
Window.Show	Shows a window.
Window.SetDisplayFormat	Sets a window's integer value display format.
Window.ShowFullScreen	Toggles main window full screen mode.
Window.WaitForUpdateComplete	Waits until all debug windows have completed updating following a change of the program execution point.

7.8.25 Watch Actions

Actions affiliated with the Watched Data Window.

Action	Description
Watch.Add	Adds an expression to the Watched Data Window
Watch.Insert	Inserts an expression into the Watched Data Window
Watch.Remove	Removes an expression from the Watched Data Window
Watch.Quick	Shows an expression within the Quick Watch Dialog